

Calculus A

Chapter 1: Functions

Section 1.3: Trigonometric Functions

Main Lecture Notes

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MATHA 1301

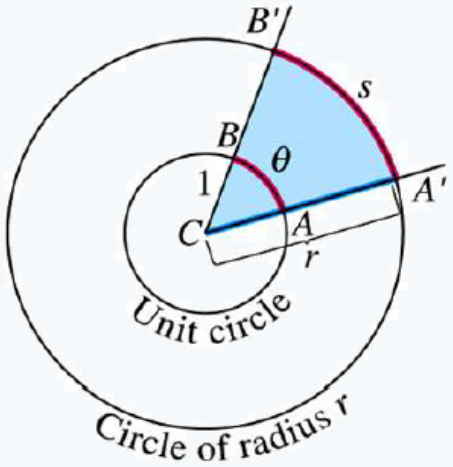
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Reference: Thomas' Calculus, 12th Edition

Trigonometric Functions

Definition

The number of radians in the central angle $A'CB'$ within a circle of radius r is defined as the number of "radius units" contained in the arc s subtended by the central angle. With the central angle measuring θ radians, this means $\theta = s/r$ or $s = r\theta$.



Note

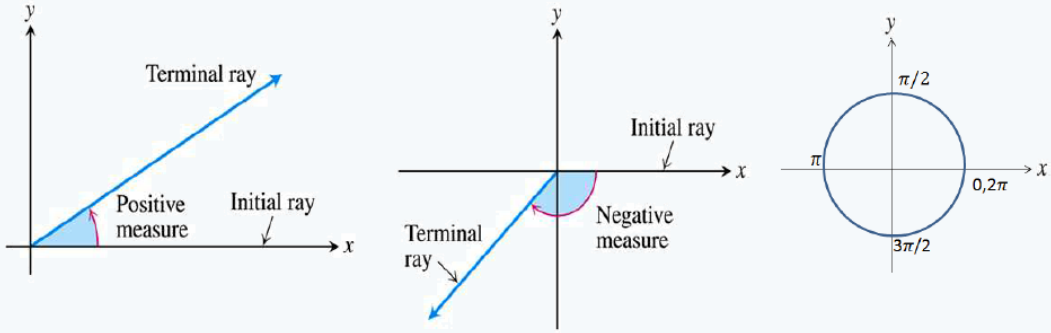
One complete revolution of the unit circle is 360° or 2π radians. Therefore π radians $= 180^\circ$ and

$$1 \text{ radian} = \frac{180}{\pi} \approx 57.3^\circ \text{ or } 1^\circ = \frac{\pi}{180} \approx 0.017 \text{ radians}.$$

Trigonometric Functions

Definition

An angle in the $xy-$plane is in standard position if its vertex lies at the origin and its initial ray lies along the positive $x-$axis. Angles measured counterclockwise from the positive $x-$axis are assigned positive measures; angles measured clockwise are assigned negative measures.



Example

A central angle in a circle of radius 8 is subtended by an arc of length 10π. Find the angle's radian in degree measures.

Solution:

$$\theta = 225^\circ$$

Trigonometric Functions

Example 1

Convert the following radians to degrees:

$$1.\ \frac{\pi}{4}.\ 2.\ \frac{\pi}{6}.\ 3.\ \frac{\pi}{3}.\ 4.\ \frac{4\pi}{3}$$

Solution:

1. $\frac{\pi}{4} = \frac{\pi}{4}.\frac{180}{\pi} = 45^\circ$
2. $\frac{\pi}{6} = \frac{\pi}{6}.\frac{180}{\pi} = 30^\circ$
3. $\frac{\pi}{3} = \frac{\pi}{3}.\frac{180}{\pi} = 60^\circ$
4. $\frac{4\pi}{3} = \frac{4\pi}{3}.\frac{180}{\pi} = 240^\circ$

Example 2

Convert the following degrees to radians:

$$1.\ 40^\circ.\ 2.\ 120^\circ.\ 3.\ 15^\circ.\ 4.\ 270^\circ$$

Solution:

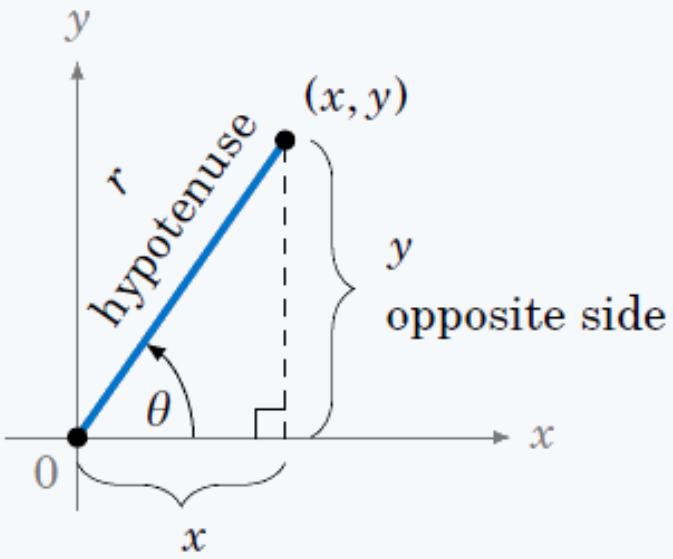
1. $40^\circ = 40.\frac{\pi}{180} = \frac{2\pi}{9}$
2. $120^\circ = 120.\frac{\pi}{180} = \frac{2\pi}{3}$
3. $15^\circ = 15.\frac{\pi}{180} = \frac{\pi}{12}$
4. $270^\circ = 270.\frac{\pi}{180} = \frac{3\pi}{2}$

Definition: Right Triangle Definition

Let $0 < \theta < \frac{\pi}{2}$ using the right triangle we define the six trigonometric functions as follows

$$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{y}{r} \qquad \cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{x}{r} \qquad \tan\theta = \frac{\text{opposite}}{\text{adjacent}} = \frac{y}{x}$$

$$\csc\theta = \frac{\text{hypotenuse}}{\text{opposite}} = \frac{r}{y} \qquad \sec\theta = \frac{\text{hypotenuse}}{\text{adjacent}} = \frac{r}{x} \qquad \cot\theta = \frac{\text{adjacent}}{\text{opposite}} = \frac{x}{y}$$



Trigonometric Functions

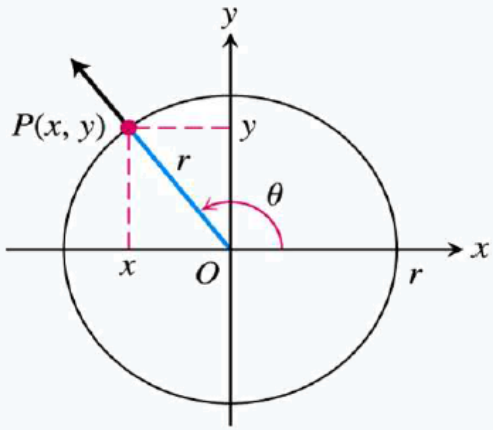
Definition: Circular Definition

We define the six trigonometric functions for any angle θ by first placing the angle in standard position in a circle of radius r. Then define the trigonometric functions in terms of the coordinates of the point $P(x, y)$ where the angle's terminal ray intersects the circle as follows:

$$\sin\theta = \frac{y}{r} \quad \cos\theta = \frac{x}{r}$$

$$\sec\theta = \frac{r}{x} \quad \csc\theta = \frac{r}{y}$$

$$\tan\theta = \frac{y}{x} \quad \cot\theta = \frac{x}{y}$$



Note 1

The coordinates of the point $P(x, y)$ on the circle $x^2 + y^2 = r^2$ can be expressed in term of θ and r as $x = r\cos\theta$ and $y = r\sin\theta$. Now, if $r = 1$ we see that $x = \cos\theta$ and $y = \sin\theta$.

Note 2

By definition above, we immediately have the trig identities:

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \quad \cot\theta = \frac{1}{\tan\theta}$$

$$\sec\theta = \frac{1}{\cos\theta} \quad \csc\theta = \frac{1}{\sin\theta}$$

Note 3

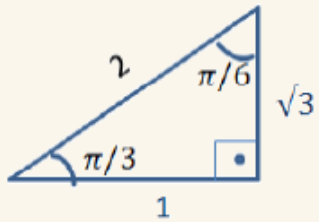
Based on the special $30^\circ - 60^\circ - 90^\circ$ and $45^\circ - 45^\circ - 90^\circ$ right triangles, we can deduce the following trig functions for the "special angles"

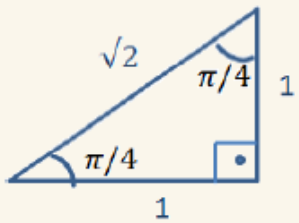
$$30^\circ = \frac{\pi}{6}, 45^\circ = \frac{\pi}{4}, \text{and } 60^\circ = \frac{\pi}{3}:$$

$$\cos\frac{\pi}{6} = \frac{\sqrt{3}}{2} \quad \cos\frac{\pi}{4} = \frac{\sqrt{2}}{2} \quad \cos\frac{\pi}{3} = \frac{1}{2}$$

$$\sin\frac{\pi}{6} = \frac{1}{2} \quad \sin\frac{\pi}{4} = \frac{\sqrt{2}}{2} \quad \sin\frac{\pi}{3} = \frac{\sqrt{3}}{2}$$

$$\tan\frac{\pi}{6} = \frac{1}{\sqrt{3}} \quad \tan\frac{\pi}{4} = 1 \quad \tan\frac{\pi}{3} = \sqrt{3}$$





Note 4

In Figure 1 we see in which quadrants or on which axes the terminal side of an angle $0° \leq \textit{theta} < 360°$ may fall.

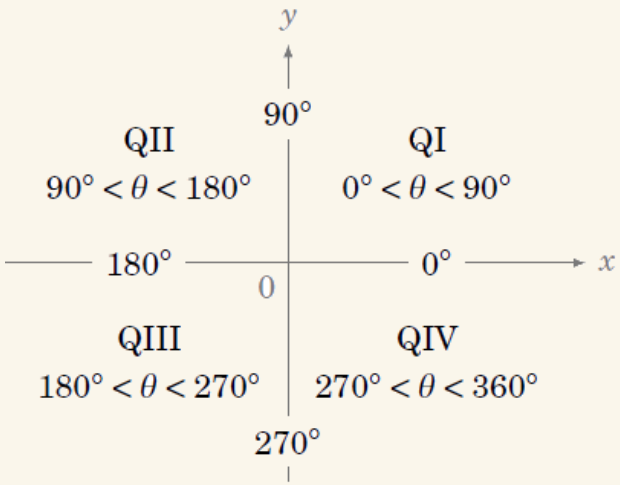


Figure: 1

Figure 2 summarizes the signs (positive or negative) for the trigonometric functions based on the angle's quadrant:

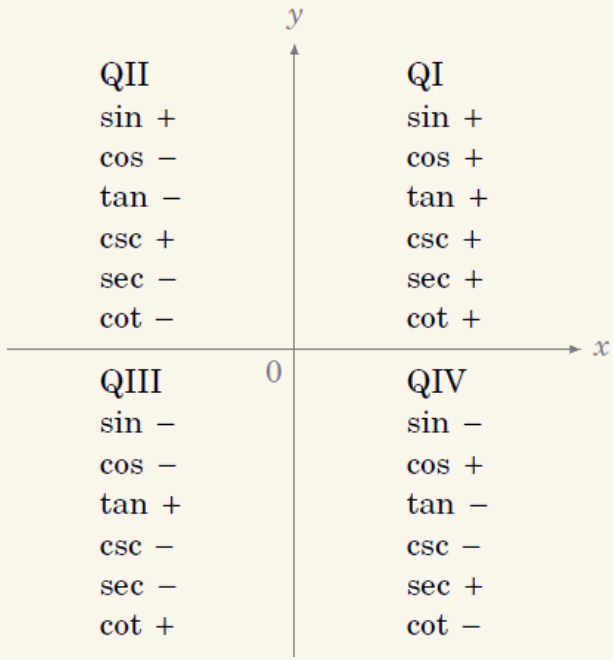


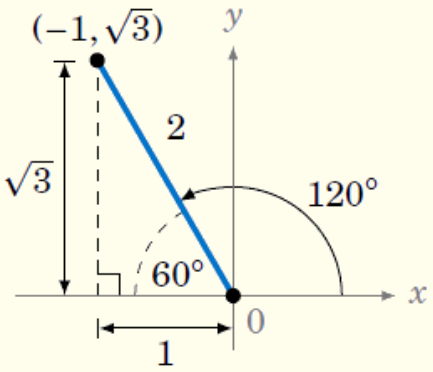
Figure: 2

Example 1

Find the exact values of all six trigonometric functions of $120°$.

Solution:

We know $120° = 180° - 60°$. We can use the point $(-1, \sqrt{3})$ on the terminal side of the angle $120°$ in *QII*, since a basic right triangle with a $60°$ angle has adjacent side of length 1, opposite side of length $\sqrt{3}$, and hypotenuse of length 2, as in the figure below. Drawing that triangle in *QII* so that the hypotenuse is on the terminal side of $120°$ makes $r = 2, x = -1$ and $y = \sqrt{3}$. Hence:



$$\sin 120° = \frac{y}{r} = \frac{\sqrt{3}}{2} \quad \cos 120° = \frac{x}{r} = \frac{-1}{2} \quad \tan 120° = \frac{y}{x} = \frac{\sqrt{3}}{-1}$$

$$\csc 120° = \frac{r}{y} = \frac{2}{\sqrt{3}} \quad \sec 120° = \frac{r}{x} = \frac{2}{-1} \quad \cot 120° = \frac{x}{y} = \frac{-1}{\sqrt{3}}$$

Example 2

Suppose that $\cos\theta = -\frac{4}{5}$. Find the exact values of $\sin\theta$ and $\tan\theta$.

Solution:

Since $\cos\theta = -\frac{4}{5}$, we can use 4 as the length of the adjacent side and 5 as the length of the hypotenuse. The length of the opposite side must then be 3. Since $\cos\theta$ is negative, so θ must be in either *QII* or *QIII* as shown in Figure 3 below.

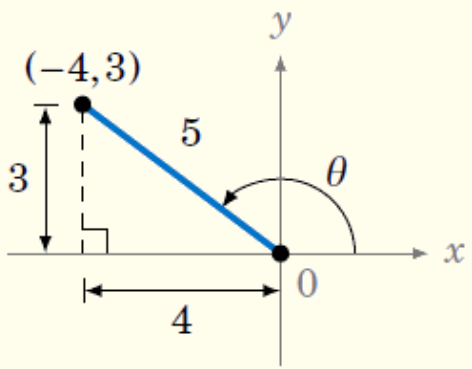


Figure 3(a)

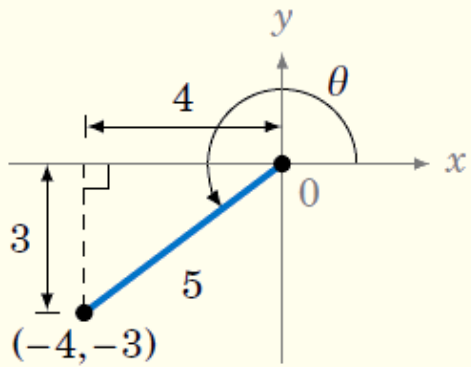


Figure 3(b)

When θ is in *QII*, we see from Figure 3(a) that the point $(-4, 3)$ is on the terminal side of θ, and so we have $x = -4, y = 3$, and $r = 5$. Thus, $\sin\theta = \frac{y}{r} = \frac{3}{5}$ and $\tan\theta = \frac{y}{x} = \frac{3}{-4}$.

Example 2 Cont..

Suppose that $\cos\theta = -\frac{4}{5}$. Find the exact values of $\sin\theta$ and $\tan\theta$.

Solution:

When θ is in *QIII*, we see from Figure 3(b) that the point $(-4, -3)$ is on the terminal side of θ, and so we have $x = -4, y = -3$, and $r = 5$. Thus, $\sin\theta = \frac{y}{r} = \frac{-3}{5}$ and $\tan\theta = \frac{y}{x} = \frac{3}{4}$. Thus, either $\sin\theta = \frac{3}{5}$ and $\tan\theta = -\frac{3}{4}$ or $\sin\theta = -\frac{3}{5}$ and $\tan\theta = \frac{3}{4}$.

Periodicity and Graphs of the Trigonometric Functions

Definition

A function $f(x)$ is periodic if there is a positive number p such that $f(x+p) = f(x)$ for every value of x. The smallest value of p is the period of f.

Note

The periods of sine, cosine, secant, and cosecant are each 2π. The periods of tangent and cotangent are both π. This leads to the trigonometric identities:

$$\sin(x+2\pi) = \sin x \quad \sin(x+2\pi) = \sin x$$

$$\sec(x+2\pi) = \sec x \quad \csc(x+2\pi) = \csc x$$

$$\tan(x+\pi) = \tan x \quad \cot(x+\pi) = \cot x$$

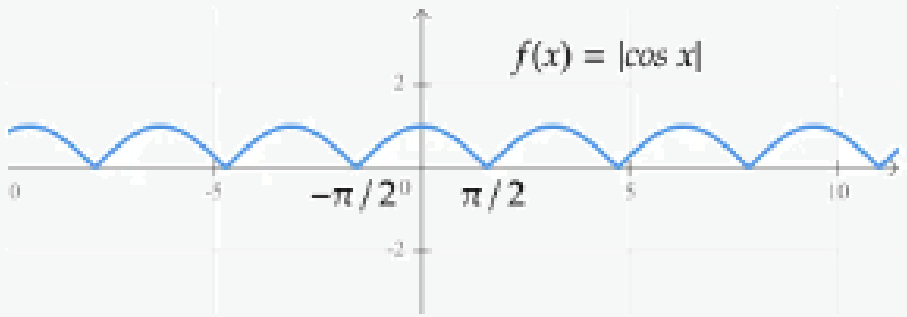
Periodicity and Graphs of the Trigonometric Functions

Note

1. If $f(x)$ has a period p, then $f(ax)$ has a period $\frac{p}{a}$.
2. If $f(x)$ has a period p, then $f(x-k)$, $f(x)+h$, $-f(x)$, $f(-x)$, and $af(x)$ each of them has the same period p.

Example 3

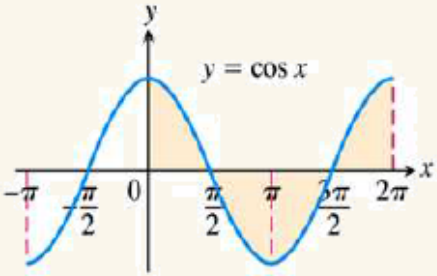
1. $f(x) = \cos(x-10) \Longrightarrow p = 2\pi$
2. $f(x) = \sin(3x) \Longrightarrow p = \frac{2\pi}{3}$
3. $f(x) = \tan(\frac{\pi x}{2}) \Longrightarrow p = \frac{\pi}{\frac{\pi}{2}} = 2$
4. $f(x) = \cot(\frac{3x}{2}) + 3 \Longrightarrow p = \frac{\pi}{\frac{3}{2}} = \frac{2}{3}\pi$
5. $f(x) = |\cos(x)| \Longrightarrow p = \pi$. Why? (see the graph of $|\cos(x)|$ below.



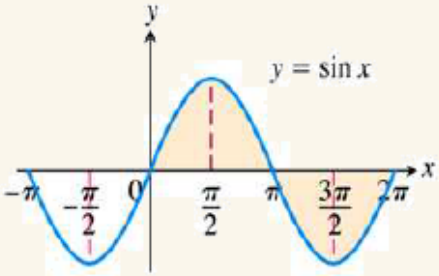
Periodicity and Graphs of the Trigonometric Functions

Note

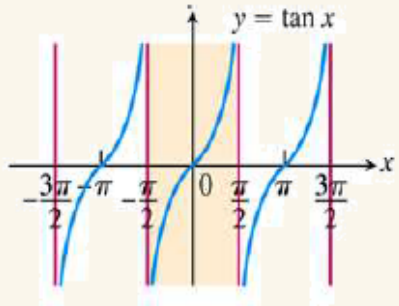
The graphs of the six trigonometric functions are as follows (the shading indicates a single period):



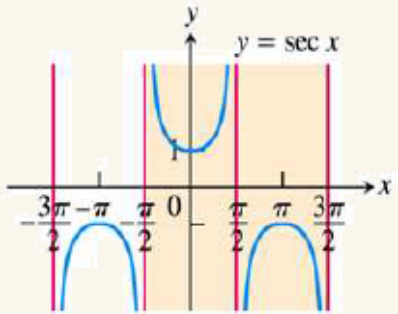
Domain: $-\infty < x < \infty$
Range: $-1 \le y \le 1$
Period: 2π



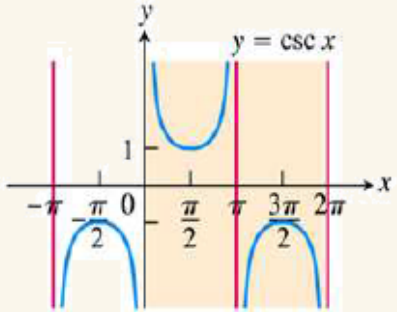
Domain: $-\infty < x < \infty$
Range: $-1 \le y \le 1$
Period: 2π



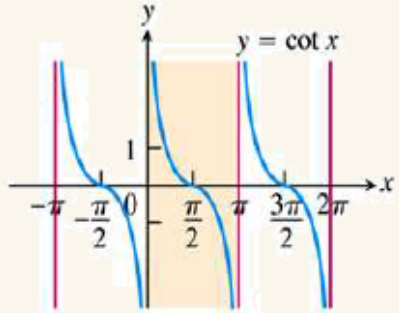
Domain: $x \neq \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \ldots$
Range: $-\infty < y < \infty$
Period: π



Domain: $x \neq \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \ldots$
Range: $y \le -1$ or $y \ge 1$
Period: 2π



Domain: $x \neq 0, \pm\pi, \pm 2\pi, \ldots$
Range: $y \le -1$ or $y \ge 1$
Period: 2π



Domain: $x \neq 0, \pm\pi, \pm 2\pi, \ldots$
Range: $-\infty < y < \infty$
Period: π

Trigonometric Identities

Pythagorean Identities:

$$\sin^2 x + \cos^2 x = 1$$

$$\tan^2 x + 1 = \sec^2 x$$

$$\cot^2 x + 1 = \csc^2 x$$

Reciprocal Identities:

$$\csc x = \frac{1}{\sin x}$$

$$\sec x = \frac{1}{\cos x}$$

$$\cot x = \frac{1}{\tan x}$$

Sum-Difference of Two Angles:

$$\sin(x \pm y) = \sin x \cos y \pm \sin y \cos x$$

$$\cos(x \pm y) = \cos x \cos y \mp \sin y \sin x$$

$$\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$$

Quotient Identities:

$$\tan x = \frac{\sin x}{\cos x}$$

$$\cot x = \frac{\cos x}{\sin x}$$

Half-Angle Formulas:

$$\sin^2 x = \frac{1 - \cos(2x)}{2}$$

$$\cos^2 x = \frac{1 + \cos(2x)}{2}$$

Double-Angle Formulas:

$$\sin(2x) = 2 \sin x \cos y$$

$$\cos(2x) = \cos^2 x - \sin^2 x$$
$$= 1 - 2\sin^2 x$$
$$= 2\cos^2 x - 1$$

Example 4

EXAMPLE . Find $\cos\left(\frac{3\pi}{4}\right)$, and $\sin\left(\frac{7\pi}{12}\right)$,

Solution:

$$\cos\left(\frac{3\pi}{4}\right) = \cos\left(\pi - \frac{\pi}{4}\right) \qquad \frac{3\pi}{4} = \pi - \frac{\pi}{4}$$

$$= \cos(\pi)\cos\left(\frac{\pi}{4}\right) - \sin(\pi)\sin\left(\frac{\pi}{4}\right) \qquad \cos(x+y) = \cos x\cos y - \sin y\sin x,$$

$$= (-1)\left(\frac{1}{\sqrt{2}}\right) - (0)\left(\frac{1}{\sqrt{2}}\right) \qquad \cos(\pi) = -1, \sin(\pi) = 0.$$

$$= -\frac{1}{\sqrt{2}}$$

$$\sin\left(\frac{7\pi}{12}\right) = \sin\left(\frac{\pi}{3} + \frac{\pi}{4}\right) \qquad \frac{7\pi}{12} = \frac{\pi}{3} + \frac{\pi}{4}$$

$$= \sin\left(\frac{\pi}{3}\right)\cos\left(\frac{\pi}{4}\right) + \sin\left(\frac{\pi}{4}\right)\cos\left(\frac{\pi}{3}\right) \qquad \sin(x+y) = \sin x\cos y + \sin y\cos x,$$

$$= \left(\frac{\sqrt{3}}{2}\right)\left(\frac{1}{\sqrt{2}}\right) + \left(\frac{1}{2}\right)\left(\frac{1}{\sqrt{2}}\right)$$

$$= \frac{1+\sqrt{3}}{2\sqrt{2}}$$

End of Section 1.3.

Thank you.!!!